

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="description" content="Data Visualisation" />
6   <meta name="viewport" content="width=device-width, initial-scale=1" />
7   <meta name="author" content="Anh Khoa" />
8   <title>LAB 6-1 D3 Interactivity - Mouse Over Effects</title>
9   <script src="https://d3js.org/d3.v7.min.js"></script>
10 </head>
11
12 <body>
13   <h1>The D3 Journey starts here...</h1>
14
15   <div style="margin-bottom: 20px;">
16     <button id="addData">Add</button>
17     <button id="removeData">Remove</button>
18   </div>
19
20   <script>
21     var w = 500;
22     var h = 200;
23     var maxValue = 30;
24
25     var dataset = [14, 5, 26, 23, 9, 12, 18, 25];
26
27     var svg = d3.select("body")
28       .append("svg")
29       .attr("width", w)
30       .attr("height", h);
31
32     const xScale = d3.scaleBand()
33       .rangeRound([0, w])
34       .paddingInner(0.1);
35
36     const yScale = d3.scaleLinear()
37       .domain([0, maxValue])
38       .range([h, 0]);
39
40     function updateChart() {
41       xScale.domain(d3.range(dataset.length));
42
43       // new bars
44       const bars = svg.selectAll("rect")
45         .data(dataset, (d, i) => i);
46
47       bars.exit()
48         .transition()
49         .duration(500)
50         .attr("x", w)
51         .remove();
52
53       const newBars = bars.enter()
54         .append("rect")
55         .attr("x", 0)
56         .attr("y", d => yScale(d))
57         .attr("width", xScale.bandwidth())
58         .attr("height", d => h - yScale(d))
59         .attr("fill", "grey");
60
61       newBars.merge(bars)
62         .transition()
63         .duration(500)
64         .attr("x", (d, i) => xScale(i))
65         .attr("y", d => yScale(d))
66         .attr("width", xScale.bandwidth())
67         .attr("height", d => h - yScale(d));
68
69       // INTERACTION
70       newBars.merge(bars)
71         .on("mouseover", function(event, d) {
72           const index = svg.selectAll("rect").nodes().indexOf(this);
73
74           d3.select(this)
75             .transition()
76             .duration(200)
77             .attr("fill", "orange");
78
79           // Hide original label
80           svg.selectAll("text.label")
81             .filter((_, i) => i2 === index)
82             .style("opacity", 0);
83
84           // Add custom tooltip
85           svg.append("text")
86             .attr("id", "tooltip")
87             .attr("x", d3.select(this).attr("x") + xScale.bandwidth() / 2)
88             .attr("y", Math.max(yScale(d) - 10, 10))
89             .attr("text-anchor", "middle")
90             .attr("font-size", "14px")
91             .attr("font-weight", "bold")
92             .attr("fill", "black")
93             .text(d);
94
95           .on("mouseout", function(event, d) {
96             const index = svg.selectAll("rect").nodes().indexOf(this);
97
98             d3.select(this)
99               .transition()
100               .duration(500)
101               .attr("fill", "grey");
102
103             d3.select("#tooltip").remove();
104
105             // Show label again
106             svg.selectAll("text.label")
107               .filter((_, i) => i2 === index)
108               .style("opacity", 1);
109           });
110
111           // LABELS
112           const labels = svg.selectAll("text.label")
113             .data(dataset, (d, i) => i);
114
115           labels.exit()
116             .transition()
117             .duration(500)
118             .attr("x", w)
119             .remove();
120
121           const newLabels = labels.enter()
122             .append("text")
123             .attr("class", "label")
124             .text(d => d)
125             .attr("x", 0)
126             .attr("y", d => Math.max(yScale(d) - 5, 10))
127             .attr("text-anchor", "middle")
128             .attr("font-size", "12px")
129             .attr("fill", "black")
130             .style("pointer-events", "none"); // Allow mouse events to pass through
131
132           newLabels.merge(labels)
133             .transition()
134             .duration(500)
135             .text(d => d)
136             .attr("x", (d, i) => xScale(i) + xScale.bandwidth() / 2)
137             .attr("y", d => Math.max(yScale(d) - 5, 10));
138         }
139
140     // Initial draw
141     updateChart();
142
143     // Button Events
144     d3.select("#addData").on("click", function () {
145       const newValue = Math.floor(Math.random() * maxValue);
146       dataset.push(newValue);
147       updateChart();
148     });
149
150     d3.select("#removeData").on("click", function () {
151       if (dataset.length > 0) {
152         dataset.shift();
153         updateChart();
154       }
155     });
156   </script>
157
158   <footer style="color: grey; margin-top: 20px;">
159     ©2024 Data Visualisation
160     Anh Khoa
161   </footer>
162 </body>
163 </html>

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